

CHAPTER 1

INTRODUCTION

1.1 Introduction

The rapid growth of **e-commerce** has transformed how consumers discover, compare, and purchase products. While online retail platforms offer convenience and accessibility, **dynamic pricing strategies** often make it difficult for users to identify the best purchasing opportunities. Product prices frequently change due to **demand fluctuations, inventory levels, promotional campaigns, and seasonal trends**.

As a result, consumers must repeatedly monitor products across multiple platforms, making the process **time-consuming, repetitive, and cognitively demanding**. Users often miss temporary discounts because continuous monitoring is impractical.

Human-Computer Interaction (HCI) focuses on designing systems that reduce user effort and improve efficiency through automation and effective feedback. To address these challenges, this project introduces **Bachat**, a smart price-tracking and automation application that enables users to monitor products, set target prices, and receive alerts when products reach their desired budget range.

1.2 Background of the Study

Online shopping has become a major part of modern consumer behavior. However, monitoring price fluctuations remains a significant challenge. Most consumers rely on **browser bookmarks, wishlists, saved links, and repeated searches** to track products of interest.

These approaches require continuous user involvement, increase **cognitive workload**, and frequently result in missed promotional opportunities. Furthermore, many existing solutions operate within individual retail ecosystems, creating **fragmented user experiences**.

The growing demand for **automation** and **personalized digital services** highlights the need for a centralized solution that simplifies price monitoring while minimizing user effort. **Bachat** addresses this need through automated tracking and intelligent notifications within a user-friendly mobile interface.

1.3 Problem Statement

Consumers who wish to purchase products at optimal prices must manually monitor multiple online stores and repeatedly check product listings for price changes. This process is inefficient, time-consuming, and heavily dependent on user memory and availability.

Existing approaches suffer from several limitations:

- **Manual monitoring requirements**
- **High cognitive workload**
- **Fragmented shopping experiences**
- **Missed promotional opportunities**
- **Limited personalization**
- **Lack of centralized tracking**

These challenges create a need for an **automated and user-centered solution** capable of continuously monitoring products and notifying users when desired pricing conditions are met.

1.4 Aim and Objectives

Aim

To design and evaluate an interactive **smart price-tracking application** that automates online product monitoring while applying **Human-Computer Interaction principles** to enhance usability and user experience.

Objectives

- Automate online product price monitoring.
- Reduce the cognitive effort associated with manual tracking.

- Provide personalized target-price notifications.
- Design an intuitive interface using **HCI principles** and **usability heuristics**.
- Evaluate and improve the system through usability testing and expert feedback.

1.5 Scope and Significance

The project focuses on the design and evaluation of a **mobile-based price-tracking application** supporting four fashion retailers:

- **Outfitters**
- **BreakOut**
- **Saya**
- **Sana Safinaz**

The system enables users to:

- Authenticate through **Magic Link Authentication**.
- Add products using URLs.
- Define target prices.
- Manage watchlists.
- Configure tracking preferences.
- Receive price-drop notifications.

The significance of **Bachat** lies in demonstrating how **HCI principles** can solve a real-world consumer problem. By automating repetitive monitoring tasks and providing meaningful feedback, the system improves efficiency, reduces user effort, and enhances the overall shopping experience.

1.6 Chapter Summary

This chapter introduced **Bachat**, highlighted the challenges of manual price monitoring, and defined the project's aim, objectives, scope, and significance. It established the foundation for developing a **user-centered and automated price-tracking solution**.

CHAPTER 2

PROBLEM ANALYSIS AND USER ANALYSIS

2.1 Introduction

A successful interactive system must address a clearly defined problem and the needs of its users. This chapter analyzes the challenges associated with online price monitoring and identifies the primary user groups targeted by **Bachat**.

2.2 Problem Analysis

Online shoppers frequently encounter difficulties when tracking product prices across multiple retail platforms. Common challenges include:

- **Manual Monitoring:** Users must repeatedly revisit product pages to identify price changes.
- **Cognitive Overload:** Managing multiple products increases mental effort and information retention requirements.
- **Fragmented Experiences:** Users switch between different websites and applications to monitor products.
- **Missed Discounts:** Flash sales and temporary promotions are often overlooked.
- **Ineffective Notifications:** Existing systems prioritize marketing campaigns over personalized price alerts.

These issues reduce efficiency and negatively impact the overall shopping experience.

2.3 Target Users

The proposed system is designed for three primary user groups:

Budget-Conscious Consumers

Users seeking affordable purchasing opportunities and maximum savings.

Young Professionals

Individuals who prefer automated solutions due to limited time availability.

Frequent Online Shoppers

Consumers who regularly purchase products online and maintain multiple wishlists.

2.4 User Requirements

Based on the identified challenges, users require:

- **Automated product monitoring**
- **Personalized price alerts**
- **Simplified onboarding**
- **Centralized watchlist management**
- **Clear system feedback**
- **Efficient interaction mechanisms**

These requirements directly influenced the design decisions implemented within **Bachat**.

2.5 System Justification

Bachat addresses the limitations of existing approaches by introducing **automation, centralized management, and user-centered interaction design**. Instead of relying on manual monitoring, the system continuously tracks products and notifies users when predefined pricing conditions are met.

This approach reduces user effort, improves efficiency, and supports better purchasing decisions.

2.6 Chapter Summary

This chapter examined the major challenges associated with online price monitoring and identified the target users of the proposed system. The findings reinforced the

need for an **automated, centralized, and user-friendly solution** capable of reducing cognitive effort while improving shopping efficiency.

CHAPTER 3

LITERATURE REVIEW

3.1 Introduction

Before developing a new solution, it is important to understand how consumers currently monitor product prices and the limitations of existing approaches. This chapter reviews common price-tracking practices and identifies the gaps that motivated the development of **Bachat**.

3.2 Existing Approaches

Browser Bookmarks and Saved Links

Many consumers save product pages for future reference.

Advantages

- Simple and easy to use.
- No additional tools required.

Limitations

- Requires manual monitoring.
- No automated notifications.
- Highly dependent on user memory.

Platform-Specific Wishlists

Retailers often provide built-in wishlist functionality.

Advantages

- Convenient product saving.
- Integrated within existing platforms.

Limitations

- Restricted to individual platforms.
- No cross-store tracking.
- Limited personalization.

Price Comparison Websites

These platforms aggregate pricing information from multiple retailers.

Advantages

- Centralized product information.
- Multi-retailer coverage.

Limitations

- Visual clutter.
- Complex interfaces.
- Poor mobile optimization.
- Advertisement-heavy layouts.

3.3 Comparative Analysis

Existing Solution	Strength	Limitation
Browser Bookmarks	Simple and accessible	Manual monitoring
Wishlists	Easy product saving	Platform dependency
Comparison Websites	Multi-retailer coverage	Complex user experience

3.4 Identified Gap and Proposed Solution

The review highlights several shortcomings in existing solutions:

- **Lack of automation**
- **Fragmented user experiences**
- **Poor mobile usability**
- **Limited personalization**
- **High cognitive workload**

To address these limitations, **Bachat** introduces:

- **Automated product tracking**
- **Personalized target-price alerts**
- **Unified watchlist management**
- **Passwordless authentication**
- **Mobile-first interface design**

By combining **automation** with **Human-Computer Interaction principles**, the system provides a more efficient, intuitive, and user-centered shopping experience.

3.5 Chapter Summary

This chapter reviewed existing price-monitoring approaches and identified their major limitations. The analysis highlighted a clear need for an **automated, centralized, and user-centered solution**, providing the foundation for the design and development of **Bachat**.

CHAPTER 4

METHODOLOGY

4.1 Introduction

This chapter describes the methodology adopted for the design and evaluation of **Bachat**. Since the project focuses on developing a user-friendly and interactive price-tracking system, the methodology is based on **Human-Computer Interaction (HCI)** principles and the **User-Centered Design (UCD)** approach.

The development process emphasized understanding user needs, designing intuitive interactions, and refining the interface through usability evaluation.

4.2 Research Approach

The project follows a **User-Centered Design (UCD)** methodology, which places users at the center of the design process. The approach focuses on identifying user requirements, developing suitable design solutions, evaluating usability, and refining the system based on feedback.

The UCD approach was selected because the success of Bachat depends heavily on **usability, efficiency, and user satisfaction**.

4.3 Development Process

The development process consisted of four phases:

Phase 1: Requirement Analysis

This phase focused on understanding the challenges associated with online price monitoring and defining the project scope.

Key activities included:

- Identifying user pain points.
- Studying shopping behaviors.
- Reviewing existing price-tracking methods.
- Defining system requirements.

Phase 2: Conceptual Design

The application's structure and interaction model were defined during this phase.

Key activities included:

- Designing user flows.
- Defining navigation architecture.
- Identifying system functionalities.
- Mapping user journeys.

User Flow:

Authentication → Onboarding → Dashboard → Product Tracking → Watchlist Management → Notifications

Phase 3: Interface Design

High-fidelity prototypes were developed using **Figma**.

The design process focused on:

- **Consistency**
- **Simplicity**
- **Visibility**
- **Error Prevention**
- **User Control**

The final prototype consisted of **15 interconnected screens** representing the complete user journey.

Phase 4: Usability Evaluation

The prototype was evaluated through:

- User Testing
- Think-Aloud Protocols
- Expert Heuristic Evaluation

The collected feedback was used to identify usability issues and improve the design.

4.4 Project Scope

To maintain a manageable project scope, the system was limited to four fashion retailers:

- Outfitters
- BreakOut
- Saya
- Sana Safinaz

These retailers were selected due to their popularity among the target audience and relevance within the local e-commerce market.

4.5 Prototype Development Tool

The prototype was designed using **Figma**, which provides:

- High-fidelity prototyping
- Interactive screen linking
- Rapid interface design
- Collaborative workflows
- Usability testing support

The tool enabled realistic simulation of user interactions before implementation.

4.6 Evaluation Techniques

Think-Aloud Protocol

Participants were asked to verbalize their thoughts while interacting with the prototype. This technique helped identify confusion, navigation difficulties, and cognitive bottlenecks.

Expert Heuristic Evaluation

An industry expert evaluated the interface using established usability principles, focusing on:

- Navigation efficiency
- Information architecture

- Interface consistency
- System feedback
- Mobile usability

The findings from this evaluation guided the redesign process discussed in Chapter 7.

4.7 Chapter Summary

This chapter presented the methodology used for the design and evaluation of **Bachat**. A **User-Centered Design** approach was adopted, involving requirement analysis, conceptual design, interface prototyping, and usability evaluation. The next chapter presents the conceptual and interface design of the proposed system.

CHAPTER 5

DESIGN

5.1 Introduction

This chapter presents the conceptual and interface design of **Bachat**. The objective was to create a user-friendly, efficient, and visually consistent application that minimizes user effort while maximizing automation and control.

The chapter discusses the interaction styles, core functionalities, user flow, and interface screens developed using Human-Computer Interaction principles.

5.2 Conceptual Design

5.2.1 Interaction Styles

The application combines multiple interaction styles:

- **Direct Manipulation** – Product cards, toggle switches, and navigation controls.

- **Form-Based Interaction** – Product entry and configuration screens.
- **Menu-Based Interaction** – Store selection dropdown menus.
- **System-Initiated Interaction** – Notifications, alerts, and status updates.

5.2.2 Core Functionalities

The main functionalities include:

- Passwordless Authentication
- Product Tracking
- Watchlist Management
- Price-Drop Alerts
- Notification Management
- Product Configuration

5.2.3 User Flow

The complete user journey is:

Splash Screen → Authentication → Onboarding → Dashboard → Add Product → Product Preview → Tracking Confirmation → Watchlist → Notifications

This workflow ensures a logical and streamlined user experience.

5.3 Interface Design

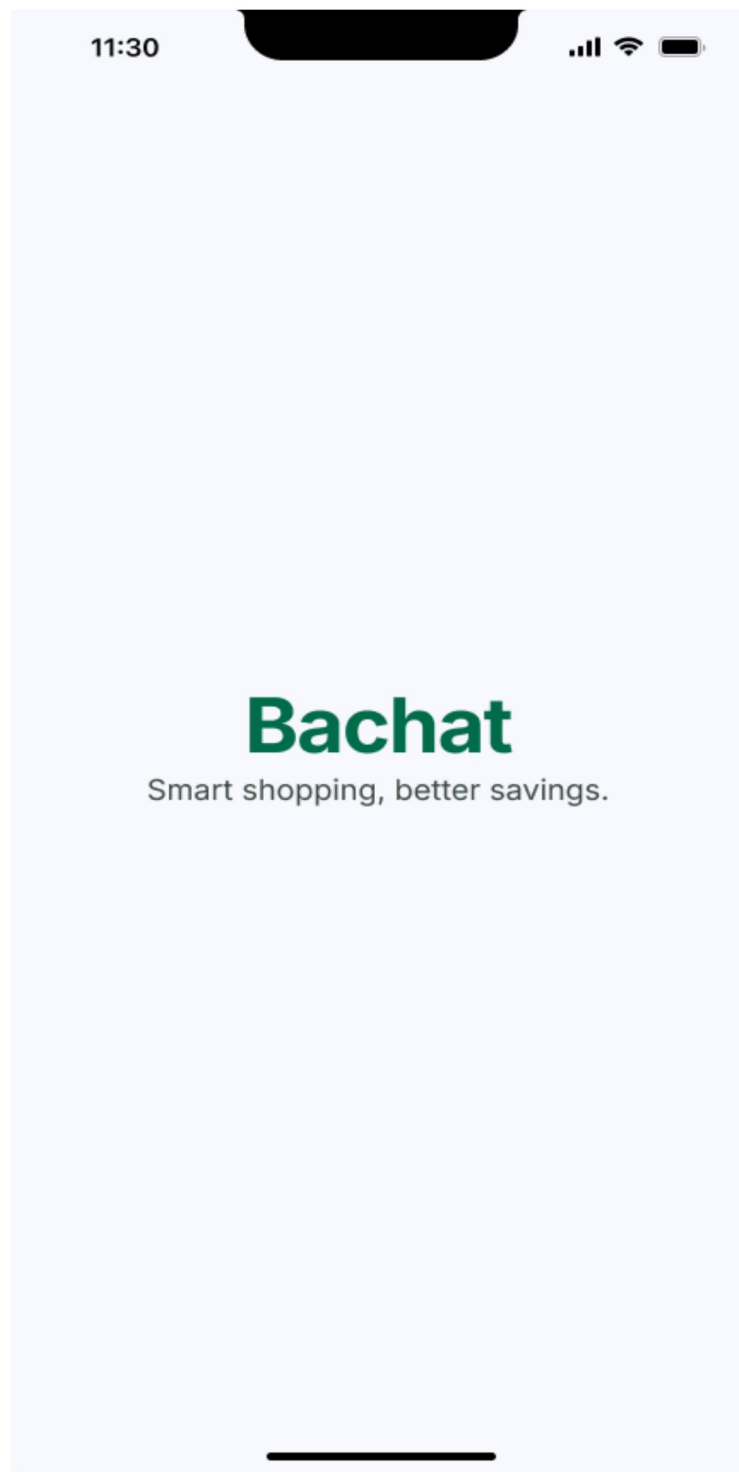
Phase 1: Authentication and Onboarding

5.3.1 Splash Screen

Purpose: Introduces the application and establishes brand identity.

Applied Heuristics:

- Visibility of System Status
- Aesthetic and Minimalist Design

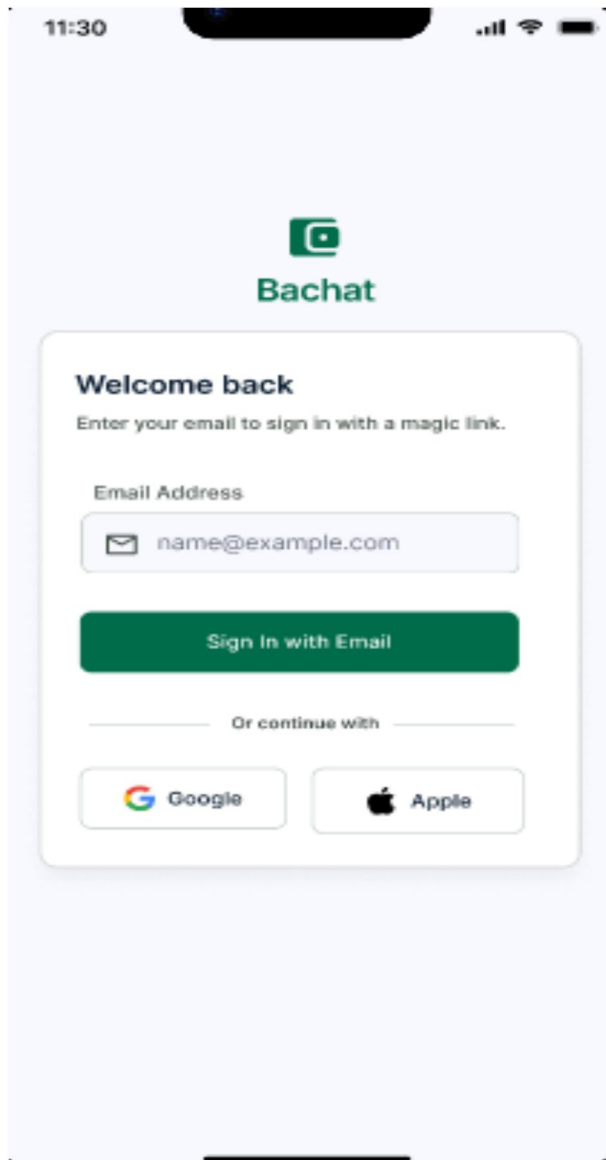


5.3.2 Passwordless Authentication

Purpose: Enables secure login through Magic Link authentication.

Applied Heuristics:

- Recognition Rather Than Recall
- Consistency and Standards

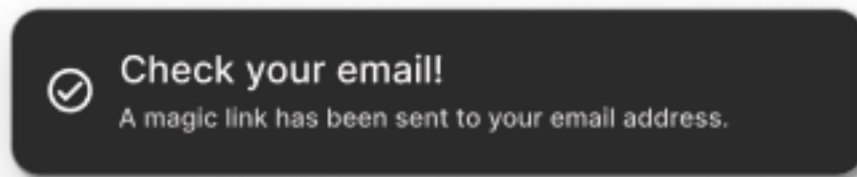


5.3.3 Verification Feedback

Purpose: Confirms successful initiation of the login process.

Applied Heuristics:

- Visibility of System Status

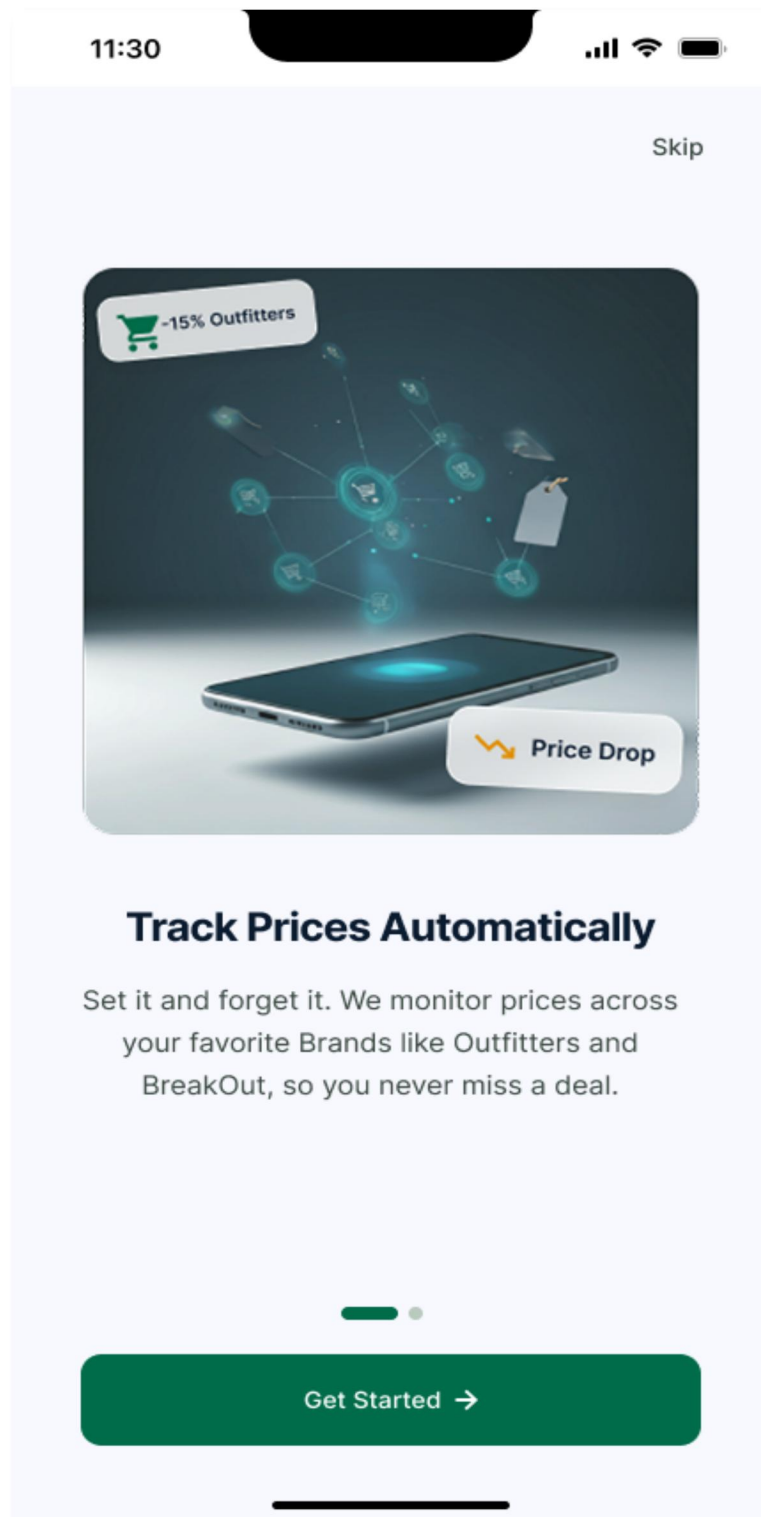


5.3.4 Onboarding – Automated Tracking

Purpose: Introduces automated price monitoring.

Applied Heuristics:

- User Control and Freedom

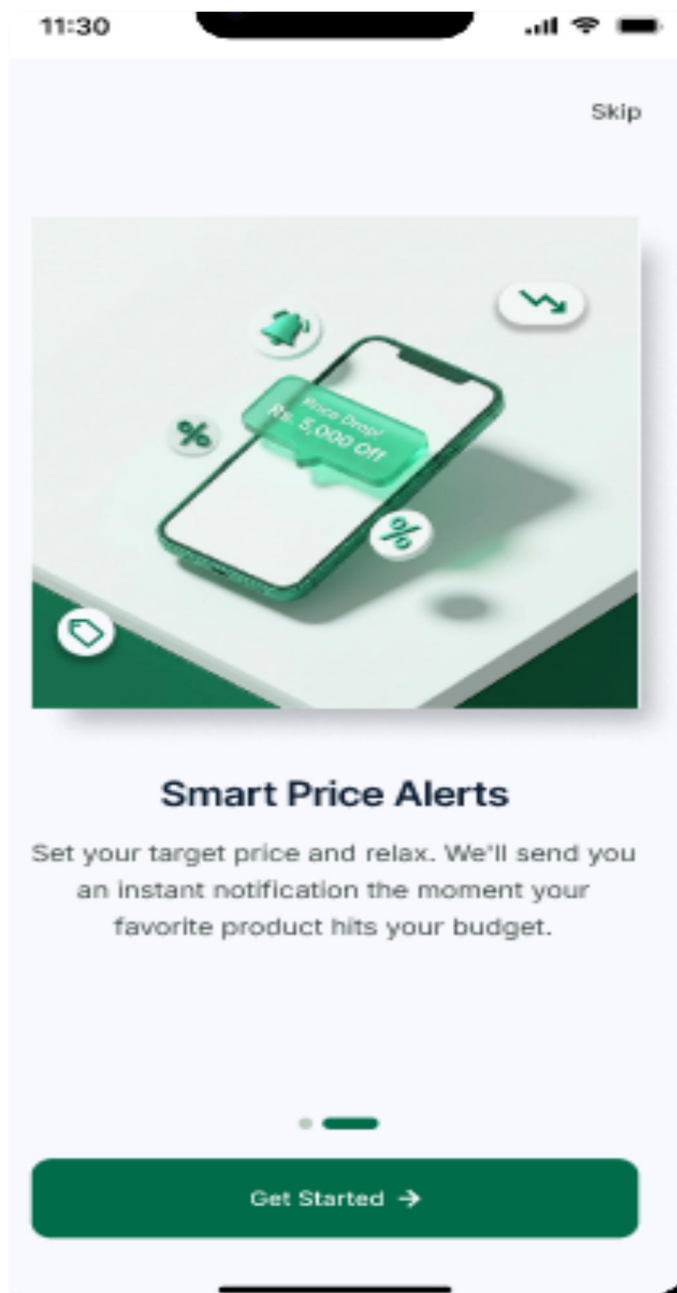


5.3.5 Onboarding – Smart Alerts

Purpose: Explains notification functionality.

Applied Heuristics:

- Consistency and Standards



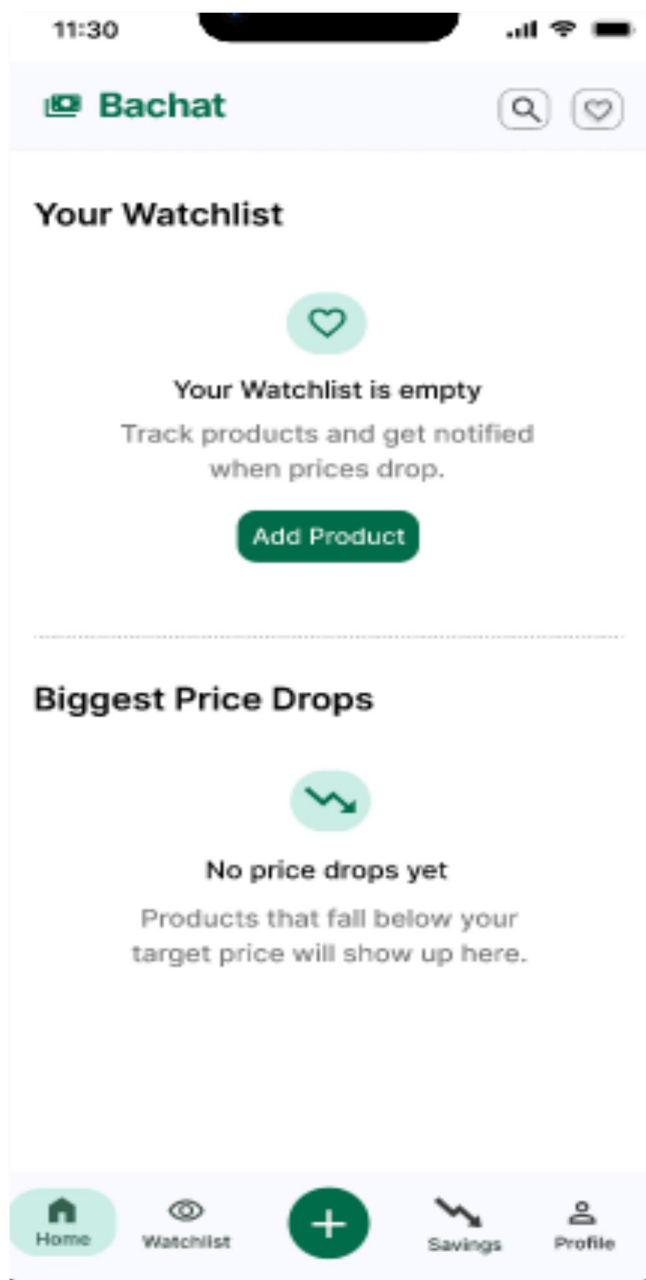
Phase 2: Dashboard Management

5.3.6 Empty Dashboard

Purpose: Guides users when no products are being tracked.

Applied Heuristics:

- Flexibility and Efficiency of Use



Phase 3: Product Acquisition

5.3.7 Add Product Screen

Purpose: Allows users to submit product URLs and define target prices.

Applied Heuristics:

- Error Prevention

11:30

Bachat

Add Product

Paste a product link below to start tracking its price automatically.

Select Store

Outfitters

Product Link

https://...

Target Price

Enter target price (PKR)

Fetch Details

Product Preview

Details will appear here once the link is processed.

Start Tracking

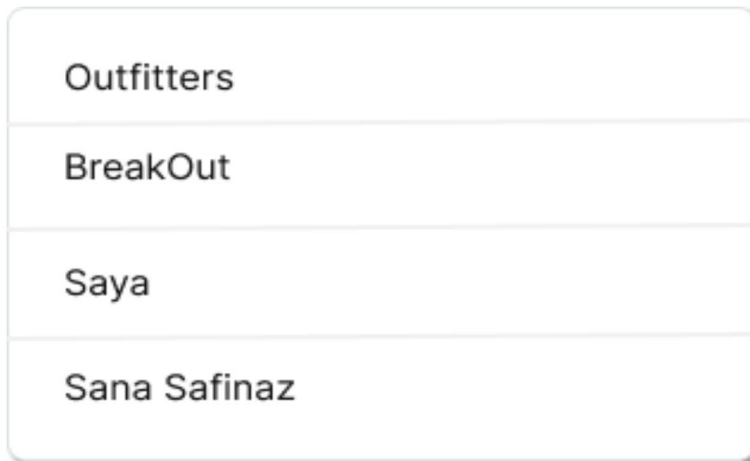
Home Watchlist Savings Profile

5.3.8 Store Selection Menu

Purpose: Enables selection of supported retailers.

Applied Heuristics:

- Match Between System and Real World

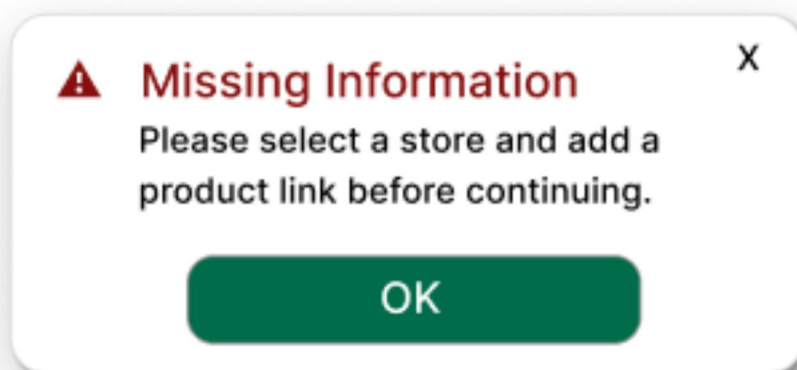


5.3.9 Validation Error Modal

Purpose: Prevents incomplete form submission.

Applied Heuristics:

- Help Users Recognize, Diagnose, and Recover from Errors



Phase 4: Product Processing

5.3.10 Product Preview

Purpose: Displays extracted product details before tracking.

Applied Heuristics:

- Visibility of System Status

The image shows a mobile application interface for 'Bachat'. At the top, the status bar shows the time 11:30 and signal strength. The app header includes the 'Bachat' logo and search and heart icons. The main section is titled 'Add Product' with the instruction 'Paste a product link below to start tracking its price automatically.' Below this, there are three input fields: 'Select Store' (a dropdown menu currently showing 'Outfitters'), 'Product Link' (a text field with 'https://...' and a clipboard icon), and 'Target Price' (a text field with the placeholder 'Enter target price (PKR)'). A green 'Fetch Details' button is positioned below these fields. The product preview section features a photograph of a man wearing 'Pleated Wide-Leg Trousers', with the product name and price 'PKR. 7,990' displayed below the image. A large green 'Start Tracking' button is at the bottom of the preview section. The bottom navigation bar contains icons for 'Home', 'Watchlist', a central '+' button, 'Savings', and 'Profile'.

5.3.11 Tracking Summary

Purpose: Allows users to review products before confirmation.

Applied Heuristics:

- User Control and Freedom

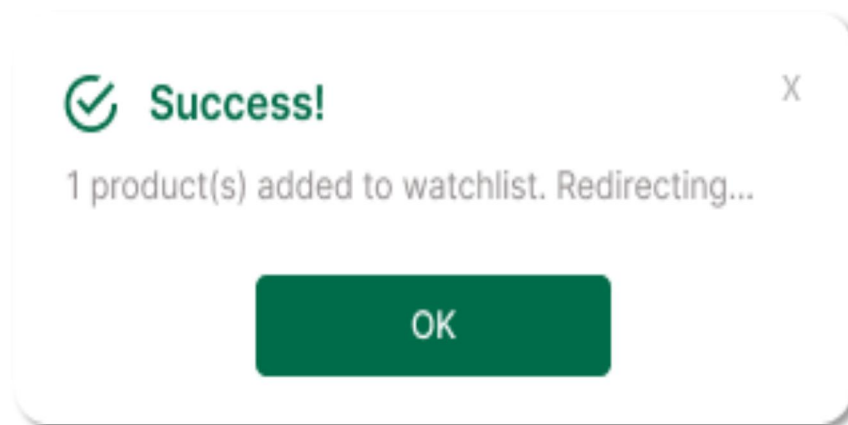
The screenshot displays the 'Bachat' app interface. At the top, the status bar shows the time as 11:30 and signal/battery icons. The app header includes the 'Bachat' logo and search/favorites icons. The main section is titled 'Add Product' with the instruction: 'Paste a product link below to start tracking its price automatically.' Below this is a form with three input fields: 'Select Store' (a dropdown menu currently showing 'Outfitters'), 'Product Link' (containing 'https://...' with a clipboard icon), and 'Target Price' (with the placeholder text 'Enter target price (PKR)'). A green 'Fetch Details' button is positioned below the form. Underneath the form is a 'Product Preview' section with a shopping bag icon and the text: 'Details will appear here once the link is processed.' A large green 'Start Tracking' button follows. The 'TRACKING (1)' section shows a single product card for 'Relaxed Fit Knitted Polo' with a price of 'PKR 3,849' and a 'Target: PKR 1500'. The card includes a product image and a close button (X). At the bottom of the tracking section is a green 'Add to Watchlist' button. The app's bottom navigation bar contains icons for 'Home', 'Watchlist', a central '+' button, 'Savings', and 'Profile'.

5.3.12 Success Confirmation

Purpose: Confirms successful activation of tracking.

Applied Heuristics:

- Visibility of System Status



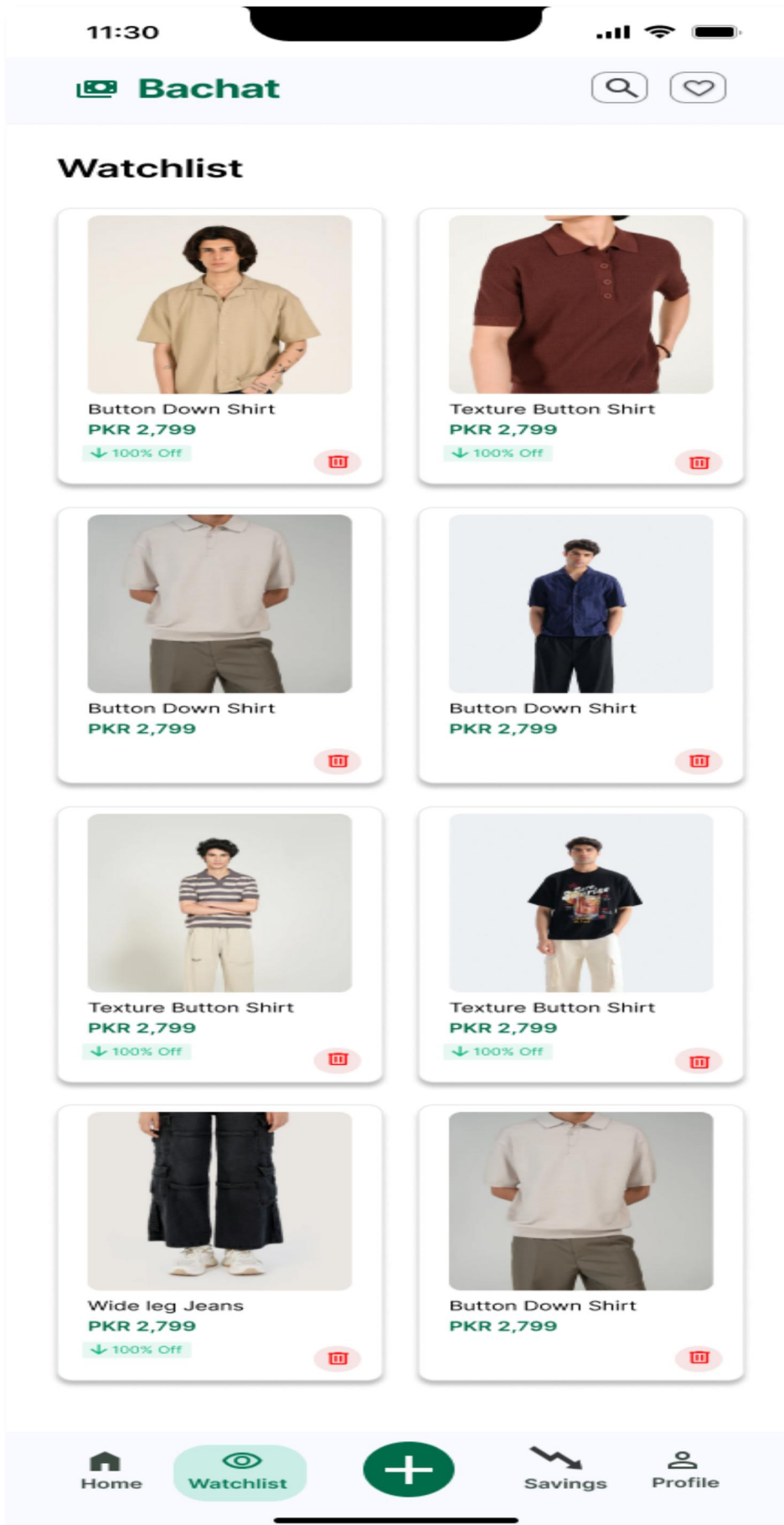
Phase 5: Watchlist Management

5.3.13 Active Watchlist

Purpose: Displays all actively tracked products.

Applied Heuristics:

- Aesthetic and Minimalist Design

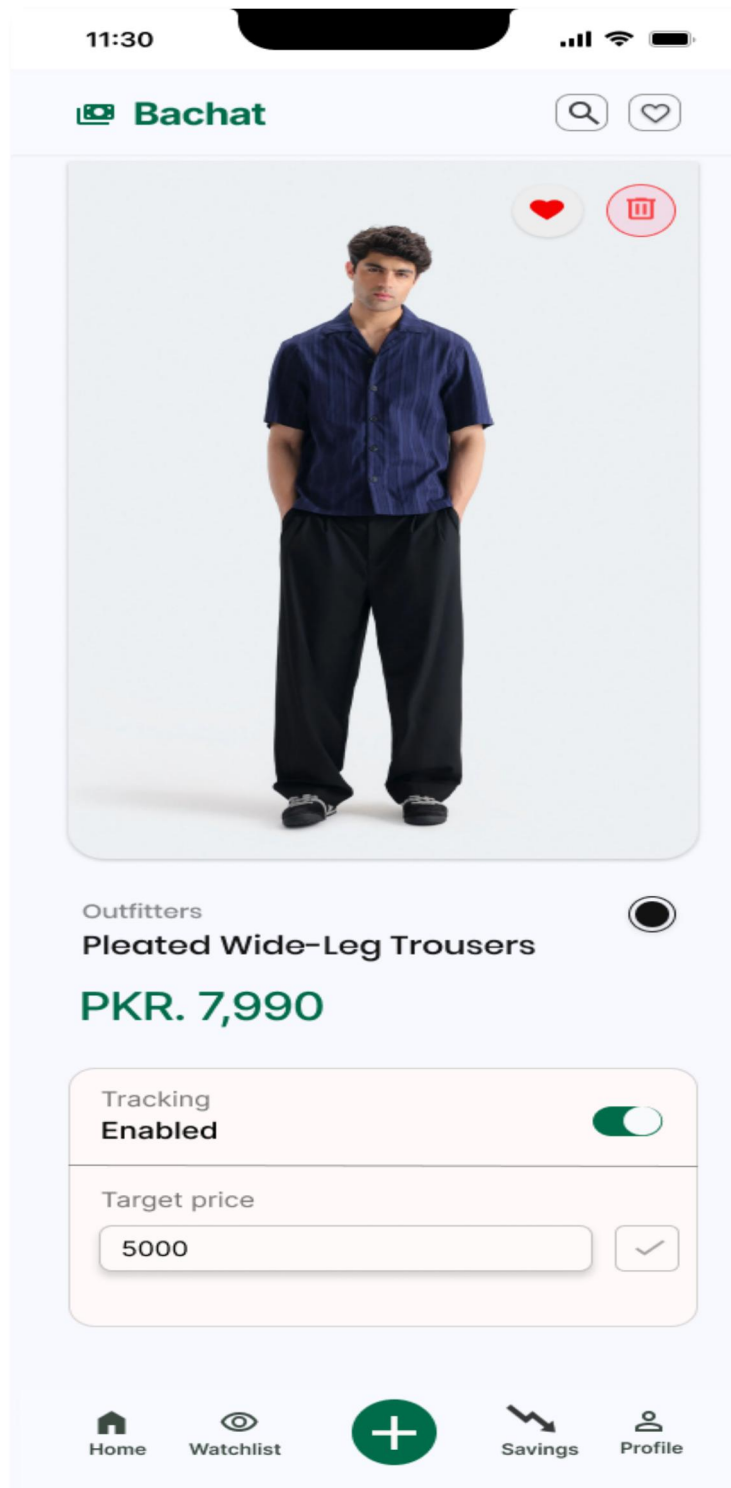


5.3.14 Product Detail Screen

Purpose: Allows users to manage tracking settings.

Applied Heuristics:

- Flexibility and Efficiency of Use

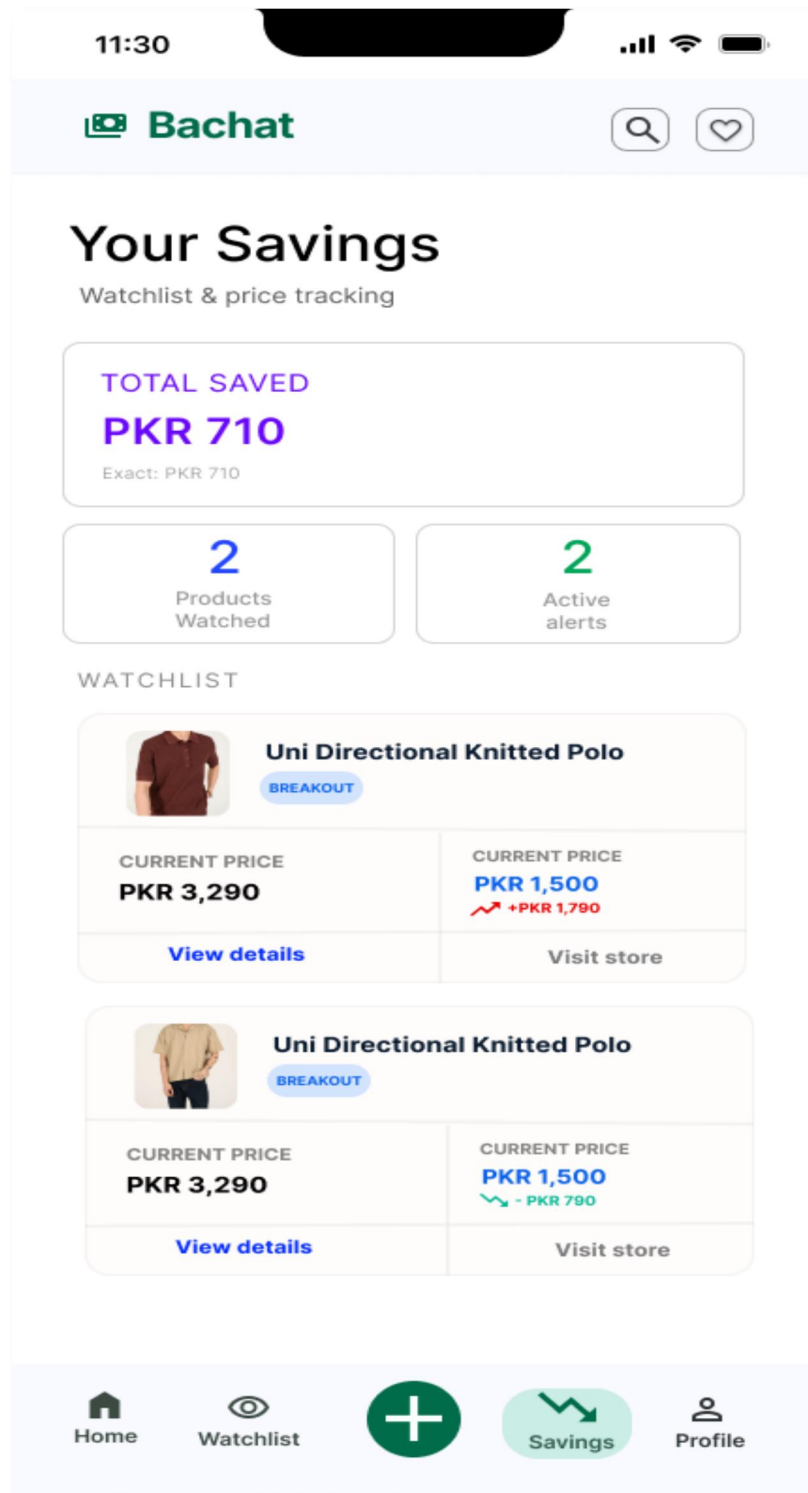


5.3.15 Savings

Purpose: Enables the user to see it's savings.

Applied Heuristics:

- User Control and Freedom



5.4 HCI Principles Applied

The interface design consistently applies Nielsen's usability heuristics:

- **Visibility of System Status**
- **Match Between System and Real World**
- **User Control and Freedom**
- **Consistency and Standards**
- **Error Prevention**
- **Recognition Rather Than Recall**
- **Flexibility and Efficiency of Use**
- **Aesthetic and Minimalist Design**
- **Error Recognition and Recovery**

These principles collectively enhance usability, efficiency, and user satisfaction.

5.5 Chapter Summary

This chapter presented the conceptual and interface design of **Bachat**. A total of **15 screens** were developed to support the complete user journey, from authentication to watchlist management. The design was guided by established **HCI principles and usability heuristics**, ensuring an intuitive and user-centered experience.

CHAPTER 6

USABILITY EVALUATION

6.1 Introduction

A formative usability evaluation was conducted on the **Bachat prototype** to assess usability, navigation clarity, and interaction efficiency. The goal was to identify usability issues before final refinement and ensure alignment with **HCI principles and User-Centered Design (UCD)**.

6.2 Evaluation Approach

A **Formative Usability Evaluation** was used, combining:

- **Think-Aloud Testing** (target users)
- **Expert Heuristic Evaluation** (industry professional)

Evaluation focused on task performance across:

Authentication, product tracking, watchlist usage, and notifications.

6.3 Participants & Tasks

Participants included **target users (online shoppers)** and an **industry expert**.

Core tasks:

- Login via Magic Link
- Add product via URL
- Set target price
- Manage watchlist
- Access product details
- Identify notification behavior

6.4 Key Findings

High Severity – Information Architecture Redundancy

- Overlap between **Watchlist** and **Savings**
- Users confused about system structure
- Violates:
 - **Flexibility & Efficiency of Use**
 - **Aesthetic & Minimalist Design**

Impact: Increased cognitive load and navigation inefficiency.

Medium Severity – Notification Disconnect

- Notification settings existed but no visible notification center
- Users unsure how alerts are delivered

Violates:

- **Visibility of System Status**
- **Match Between System and Real World**

Impact: Reduced user trust and system transparency.

Low Severity – Onboarding Confusion

- Users unclear about first actions on dashboard
- Limited contextual guidance

Violates:

- **Help & Documentation**

Impact: Reduced learnability for first-time users.

Additional Observation – Platform Consistency

- Interface resembled a web app inside mobile view
- Needed stronger adherence to **mobile-native design standards**

6.5 Chapter Summary

The evaluation identified critical usability issues in **navigation structure, notification visibility, and onboarding clarity**. These findings directly informed the redesign phase to improve usability and system coherence.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 Introduction

Based on usability findings, the Bachat interface was refined to improve **navigation efficiency, clarity, and user experience**.

7.2 Key Redesigns

1. Quick Start Onboarding Tutorial

Added:

- Introduced a **“Learn Bachat in 30 Seconds”** onboarding screen to provide immediate user guidance.
- Integrated a tutorial video placeholder demonstrating the application's core functionality and potential savings.
- Included a **“What You'll Master”** section highlighting key features such as **Price Tracking** and **Price Drop Alerts**.
- Provided **Watch Tutorial**, **Get Started**, and **Skip** options to support different user preferences.

Benefits:

- Accelerated user learning and feature understanding.
- Improved first-time user experience and onboarding efficiency.
- Established clear expectations regarding application functionality.
- Enhanced user control and flexibility during the onboarding process.

2. Navigation and Profile Architecture

Added:

- Implemented a structured **five-tab navigation system** consisting of **Home**, **Watchlist**, **Add Product**, **Savings**, and **Profile**.
- Developed a dedicated **Profile Management Screen** for account information and preferences.
- Centralized account-related functions, including **Notification Settings**, **Tutorial**

Access, and Account Management.

- Added clearly distinguishable **Log Out** and **Delete Account** controls for improved account accessibility.

Benefits:

- Reduced cognitive load through organized feature grouping.
- Improved navigation efficiency and overall usability.
- Strengthened information architecture by separating core and secondary functions.
- Enhanced accessibility of frequently used account settings and preferences.

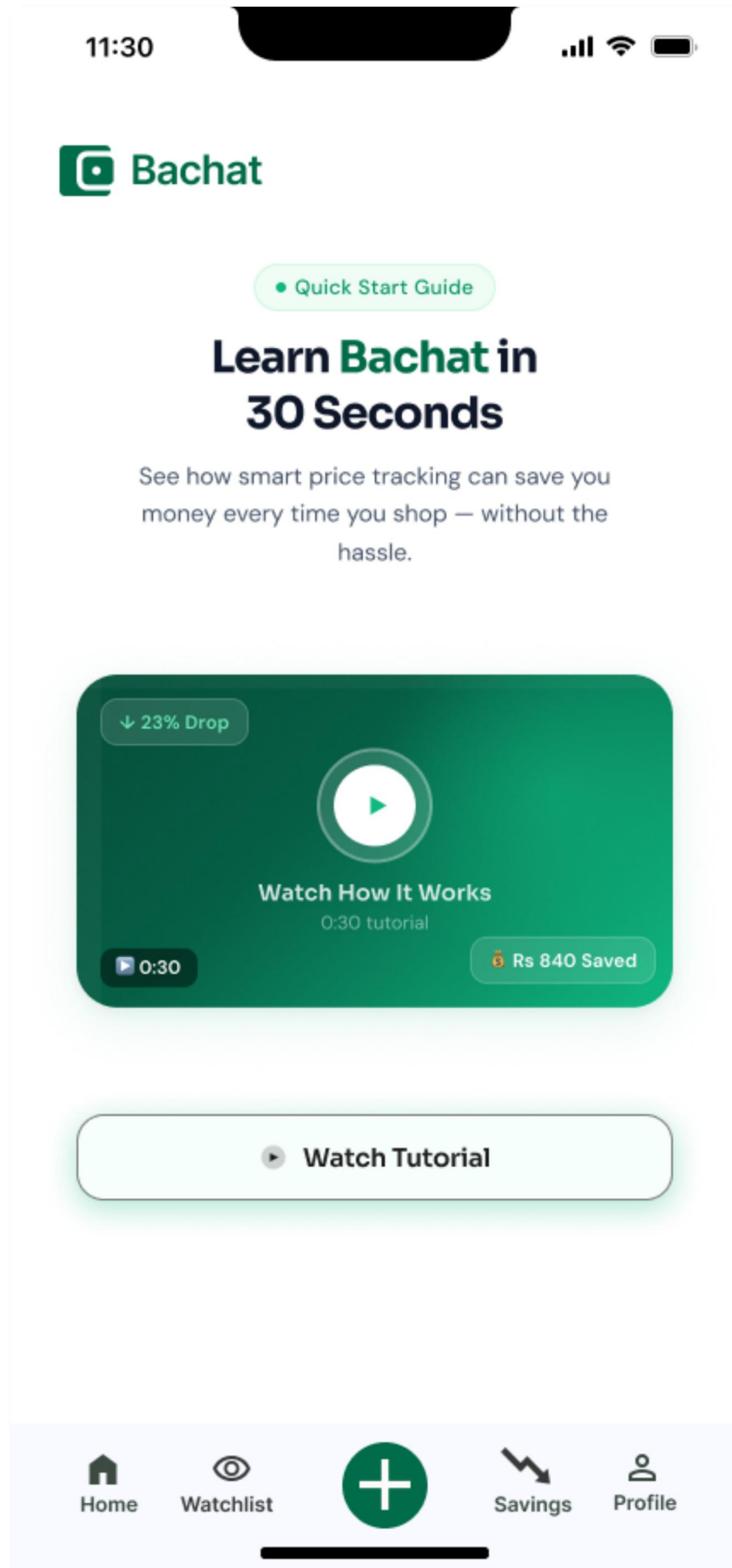
3. Product Discovery and Price Drop Feed

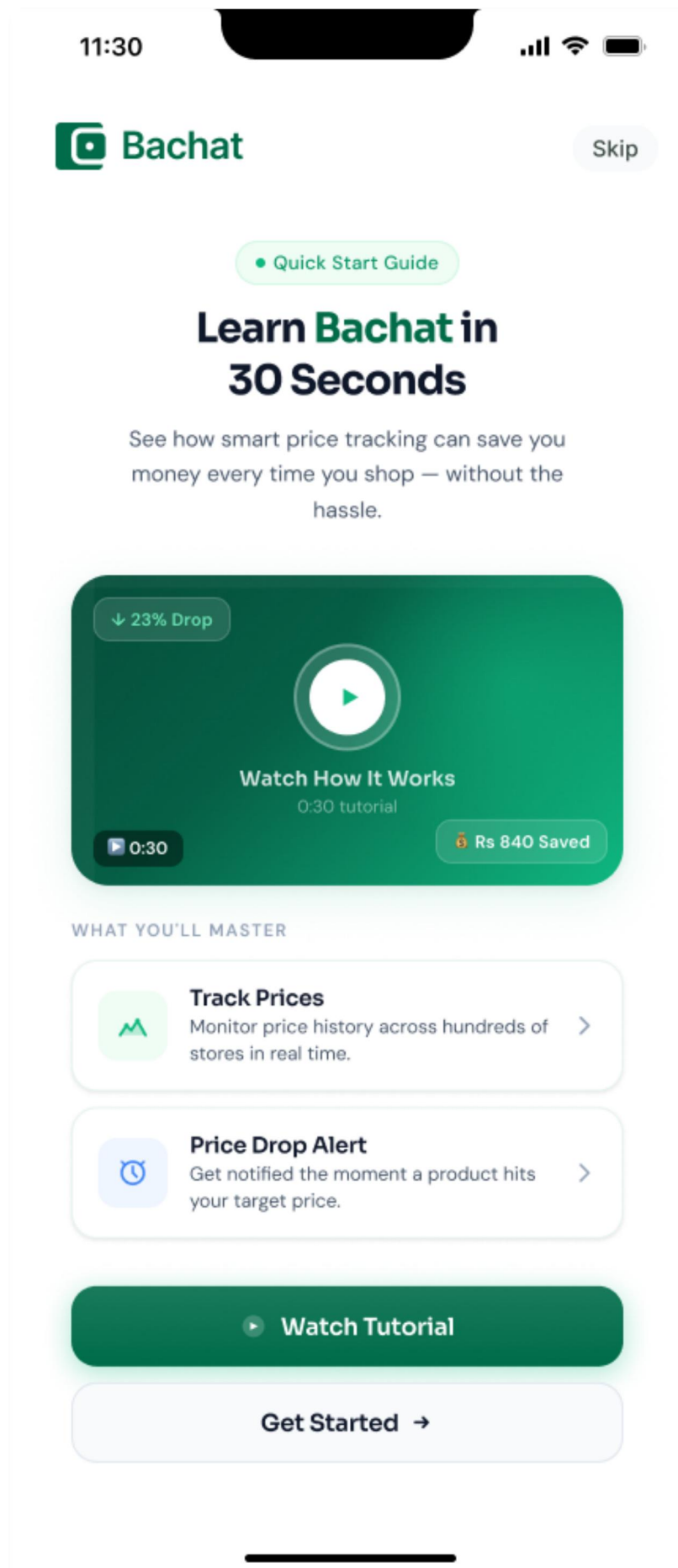
Added:

- Introduced a utility toolbar containing **Search**, **Favorites**, and **Help/Tutorial** shortcuts.
- Added a **Recently Added Products** carousel for quick access to tracked items.
- Implemented a **Biggest Price Drops** section highlighting active discounts and savings opportunities.
- Displayed product comparisons using original prices, discounted prices, and promotional badges.
- Included retailer-specific product information to improve tracking transparency.

Benefits:

- Improved visibility of tracked products and active discounts.
- Increased user trust through transparent price comparison mechanisms.
- Provided continuous feedback regarding system activity and savings opportunities.
- Enhanced product discoverability and user engagement.
- Improved awareness of real-time price reductions and promotional offers.





11:30

Profile

Your Name

name@example.com

FULL NAME

Enter your name

EMAIL ADDRESS

name@example.com

Email cannot be changed

Save Changes

ACCOUNT

Watch Tutorial >

Notification Settings >

Delete Account >

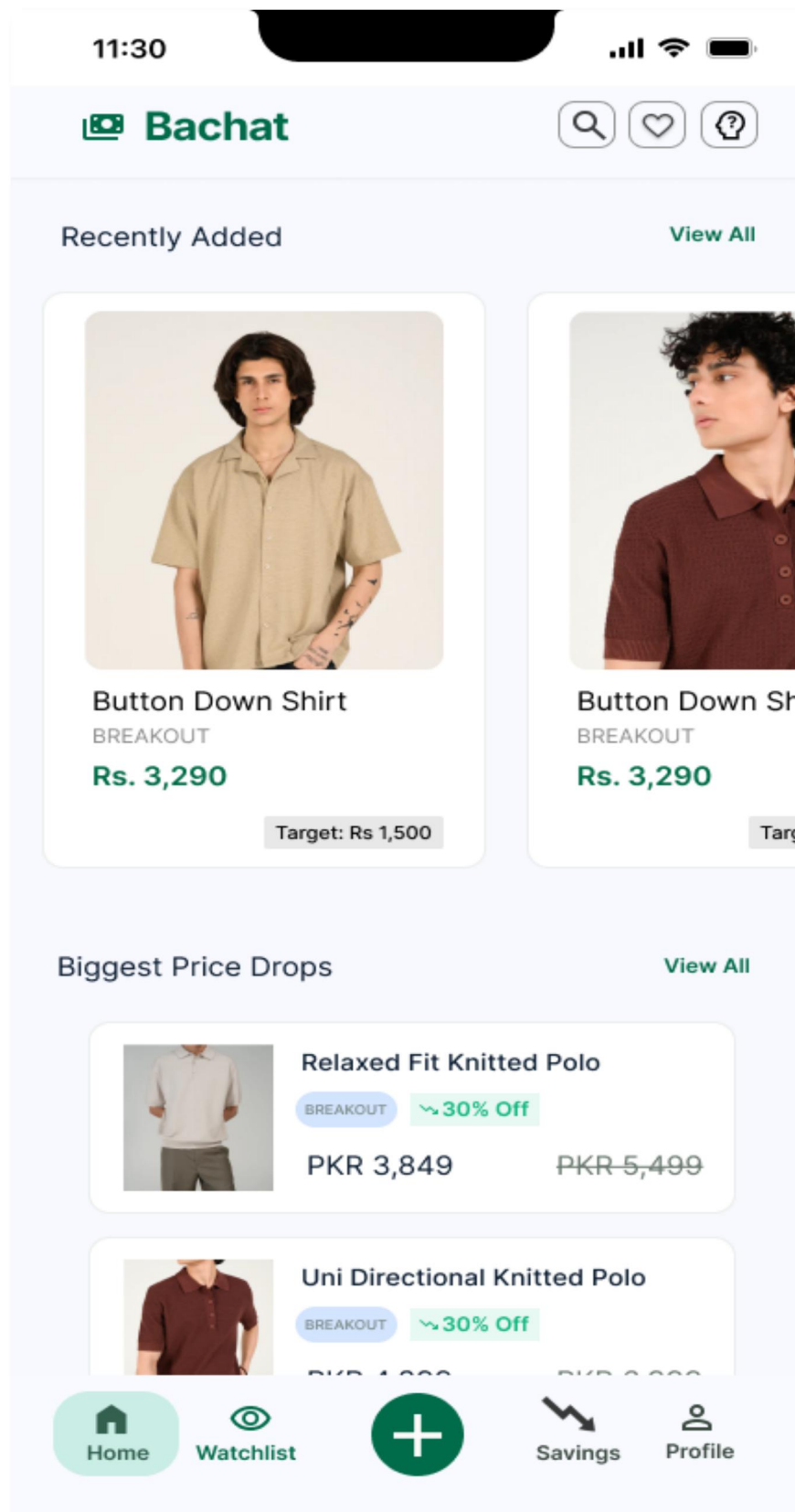
Permanently remove your data

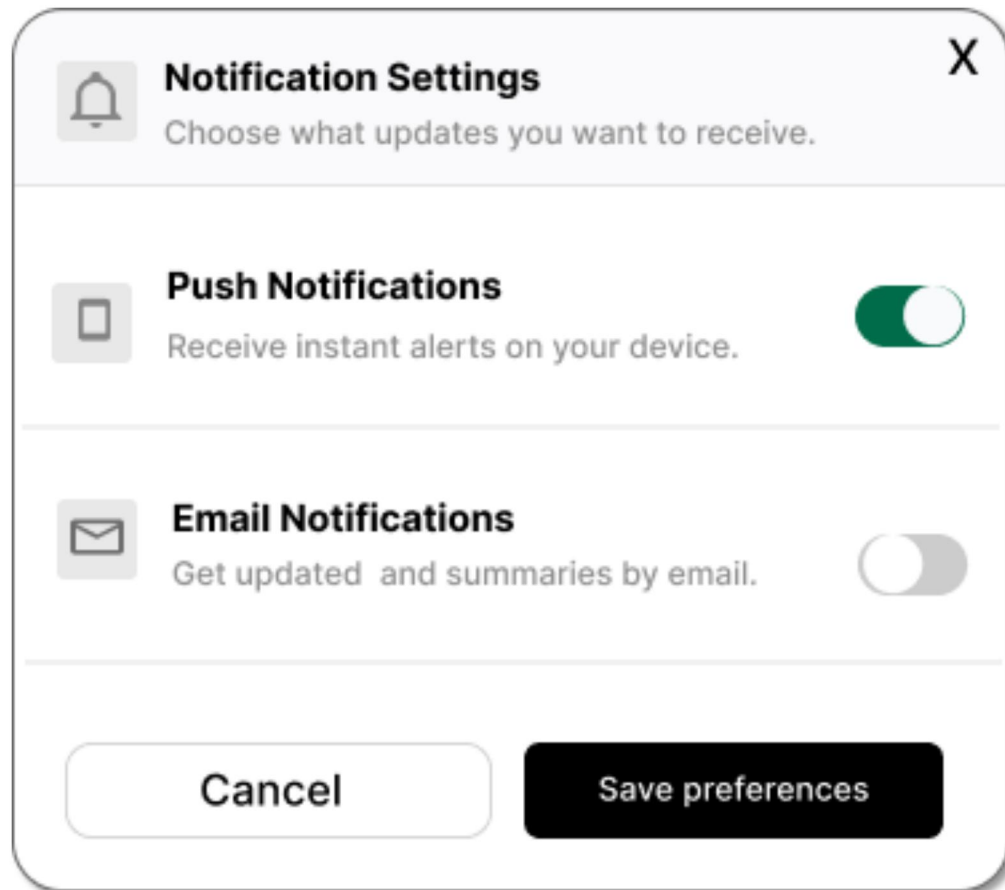
Home

Watchlist

Savings

Profile





7.3 Discussion

The redesign confirms that usability issues often emerge despite visually strong interfaces. Evaluation-driven design significantly improved:

- Navigation structure
- System feedback clarity
- Learnability
- Interaction efficiency

7.4 Retained Strengths

- Strong **error prevention mechanisms**
- Clean visual hierarchy
- Passwordless authentication flow
- Consistent branding
- Clear interactive affordances

7.5 Chapter Summary

The refined design successfully resolved major usability issues identified during evaluation, resulting in a more **cohesive, efficient, and user-centered system**.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project successfully designed and evaluated **Bachat**, a smart price-tracking system developed using **HCI principles and UCD methodology**.

The system automates price monitoring, reducing manual effort and cognitive load while improving shopping efficiency through:

- Product tracking
- Target price alerts
- Watchlist management
- Automated notifications

Usability evaluation ensured iterative refinement, significantly improving system usability and structure.

8.2 Achievement of Objectives

All project objectives were achieved:

- HCI principles applied
- Conceptual model developed
- UI designed and prototyped
- Usability evaluation conducted
- Iterative improvements implemented

8.3 Future Enhancements

Future improvements may include:

- Expanded retailer support
- AI-based price prediction
- Personalized recommendations
- Analytics dashboard (price history & savings)
- Real-time push notifications
- Cross-platform synchronization
- Social sharing of deals

8.4 Final Remarks

Bachat demonstrates that effective interface design is not only visual but **behavior-driven and usability-focused**. By combining automation with HCI principles, the system provides a practical, efficient, and user-centered solution to modern e-commerce price tracking challenges.